



## Sudarshan Jha

Roll No.: 20247051

B.Tech.

Production and Industrial Engineering

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## EDUCATION

### •Motilal Nehru National Institute Of Technology

2024-Present

B.Tech Production and Industrial Engineering

CPI: 9.15(Till 4th Sem)

### •Public School Darbhanga

2024

Central Board Of Secondary Education, Bihar

Percentage: 93.5

### •Madonna English School

2022

Indian Certificate of Secondary Education, Bihar

Percentage: 97.2

## PERSONAL PROJECTS

### •TripMate AI – Multi-Agent Travel Planner

2026

Agentic AI system that converts natural-language trip requests into full itineraries | [GitHub](#) | [Live](#)

- Tools & technologies used: Python, LangGraph, LangChain, FastAPI, PostgreSQL, Groq, Tavily API, Aviation-Stack API
- Designed a multi-agent LangGraph workflow with dedicated agents for flight search, hotel research, itinerary generation, and response formatting, orchestrated through a stateful graph
- Implemented conversation state persistence using PostgreSQL checkpointing, allowing multi-turn context to be retained across requests
- Deployed a production FastAPI backend with a live web interface, integrating real-time flight and travel data via external APIs

### •SnapClass – AI Attendance System

2026

AI web app automating classroom attendance via face recognition and voice recognition | [GitHub](#) | [Live](#)

- Tools & technologies used: Python, Streamlit, OpenCV, face\_recognition (dlib), Supabase
- Architected a multi-role system (Teacher/Student) with session-state-driven screen routing; teachers capture a single classroom photo to auto-mark attendance while students self-enroll via shareable join-code URLs
- Integrated Supabase for real-time storage of student rosters, attendance logs, and secure authentication across sessions

### •Text Summarizer API

2024 – Present

Abstractive NLP API using T5-Small Transformer for real-time text summarization | [GitHub](#)

- Tools & technologies used: Python, FastAPI, PyTorch, HuggingFace Transformers (T5-Small), Pydantic
- Built a REST API with a POST /summarize endpoint that performs abstractive summarization on long-form text using the T5-Small model ( 60M parameters), enabling real-time inference
- Handled input validation via Pydantic schemas and served the model efficiently using Uvicorn/FastAPI for low-latency responses

### •Thyroid Anomaly Detection

2024 – Present

Unsupervised ML pipeline for medical anomaly detection using clustering and tree-based methods | [GitHub](#)

- Tools & technologies used: Python, Scikit-learn, Pandas, NumPy, Matplotlib
- Applied PCA for dimensionality reduction, DBSCAN for density-based clustering, and Isolation Forest for outlier detection on a medical thyroid dataset
- Visualized cluster boundaries and anomaly scores to interpret model outputs without labeled data

## TECHNICAL SKILLS AND INTERESTS

**Languages:** Java, Python, JavaScript

**Developer Tools:** Git, GitHub, Postman, VS Code

**Frameworks:** FastAPI, Node.js, React, Express.js, Streamlit, Flask

**Cloud/Databases:** MySQL, Vector Databases(ChromaDB, Pinecone)

**Soft Skills:** Problem Solving, Analytical Thinking

**Coursework:** Data Structures and Algorithms, DBMS

**Areas of Interest:** Artificial Intelligence, Machine Learning, Deep Learning, LLM, RAG, Generative AI

## POSITIONS OF RESPONSIBILITY

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- Mentor**, Student Mentorship Program(SMP), MNNIT Allahabad *2025-Present*
  - Mentoring junior students regarding academics, campus resources, and extracurricular opportunities.
  - Conducting regular one-on-one interactions to address academic and personal challenges.
  - Facilitating smooth transition of mentees into college life through guidance and support.
- Technical Team Member(Software)**, Aeromodelling Club, MNNIT Allahabad, *Position tenure*
  - Contributing to software related projects and technical development within the club.

## ACHIEVEMENTS

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- Data Structures and Algorithms** Solved 500+ problems across different platforms like Leetcode and GeeksforGeeks